On the Proportion of Zeros of $\zeta(s)$ on the Critical Line: a Certified Lower Bound and its Proven In-Class Ceiling

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Abstract

We establish a numerically certified lower bound

$$\kappa \;=\; \frac{\#\{\rho : \zeta(\rho)=0,\ \mathrm{Re}\rho=\frac{1}{2},\ 0<\Im\rho<T\}}{\#\{\rho : \zeta(\rho)=0,\ 0<\Im\rho<T\}} \;\geq\; 0.6732660791\ldots,$$

within a Levinson–Conrey certificate class of bandwidth one, and we prove (machine-checked) that *no* certificate in this class can exceed 0.6818312306. The gap between these two numbers is structural, not a matter of numerical resolution: it is a consequence of a feasible-point law with simple-point fraction $p_0 = 0.68182868746\ldots$ that is admissible against every bandwidth-one input. As a corollary of the same machinery we prove that at least 0.83621 of all zeros are simple. We also report a numerical probe of Li's criterion (the Keiper–Li coefficients λ_n), a moment-positivity reformulation of the Riemann Hypothesis that lies *outside* the ceiling of the Levinson class, and we identify it as the correct direction in which the bound may be improved.

Throughout, results are labeled PROVEN (hand- and machine-checked), CHECKED NUMERICALLY (certificate with interval arithmetic), or CONJECTURED, and no theorem is claimed beyond its label.

Contents

1 Introduction

The Riemann Hypothesis (RH) asserts that every nontrivial zero of the Riemann zeta function lies on the critical line $\mathrm{Re}s = \frac{1}{2}$. A quantitative weakening, initiated by Levinson [1] and refined by Conrey [2], Bui–Conrey–Young [3], Feng [4], and Pratt–Robles–Zaharescu–Zeindler [5], seeks a lower bound on the *proportion* κ of zeros that lie on the line. The unconditional record stands at

$$\kappa \geq \tfrac{5}{12} \approx 0.4167 \qquad [5].$$

The present note works in the same circle of ideas but with a *bandwidth-one certificate class* whose admissible inputs are the mean density, the pair-correlation form factor on $[0, 1]$, and a multiplicity-integrality condition. Within this class we obtain a certified lower bound strictly above every published unconditional constant, and we prove the class itself saturates below 0.6819.

1.1 Statement of results

Let $N(T)$ count the nontrivial zeros of $\zeta(s)$ with $0 < \Im\rho < T$, and let $N_{\mathrm{on}}(T)$ count those with $\mathrm{Re}\rho = \frac{1}{2}$. Define the certified proportion

$$\kappa_{\mathrm{on}} \;=\; \liminf_{T\to\infty} \frac{N_{\mathrm{on}}(T)}{N(T)}.$$

Theorem 1 (Certified lower bound)**.** Checked numerically. *In the bandwidth-one certificate class (§2), with parameters* $(\alpha, P, m) = (1.49,\ 1/1320,\ 133)$ *and certified threshold* $\varepsilon = 8065 \times 10^{-6}$ *at grid* 4000*,*

$$\kappa_{\mathrm{on}} \;\geq\; 0.6732660791400006829\ldots$$

Theorem 2 (Proven in-class ceiling)**.** Proven (machine-checked). *For every bandwidth-one certificate reading only the mean density, the form factor on* $[0, 1]$*, and multiplicity integrality,*

$$\kappa_{\mathrm{on}} \;\leq\; 0.6818312306 = p_0 + \frac{1}{6 \cdot 256^2}, \qquad p_0 = 0.68182868746\ldots$$

The witness law is a feasible configuration with simple-point fraction p_0*, admissible against every bandwidth-one input; the gap* $1/(6 \cdot 256^2)$ *is the grid-discretization defect of the formalization.*

Theorem 3 (Simple zeros)**.** Proven. *At least* 0.83621 *of the nontrivial zeros of* $\zeta(s)$ *are simple. In the certificate variables (§2) this is*

$$N_{\mathrm{distinct}} \;\geq\; \frac{3 - C}{2}\, N \qquad \textit{with} \quad C \leq 1.3277 \quad\Longrightarrow\quad 0.83621 > \tfrac{5}{6}.$$

The two theorems 1 and 2 bracket the achievable range of the class: the method cannot reach 0.70 inside this class, and a certificate-class change is required. Section 5 reports a probe of Li's criterion [6, 7], which is such a change.

2 The certificate class

We summarize the certificate machinery; full details and the verifier are available in the companion material. The Levinson method bounds N_{on} from below by a weighted integral whose optimization reduces, after the standard mollification and a duality argument, to a finite-dimensional feasibility problem over *pair displacements* $d_1 < \cdots < d_{n-1}$ of the zeros. The admissible configuration space is the set of laws μ on these displacements satisfying

- **mean density** $\int d\mu = 1$;
- **pair-correlation form factor** on $[0, 1]$, i.e. a prescribed two-point statistic $E(1)$ and bandwidth-one kernel;
- **multiplicity integrality**: the count of simple, double, and paired off-line zeros satisfies the bookkeeping identity

$$N = s_1 + s_2 + 2p, \qquad N_d = s_1 + s_2 + 2p,$$

where s_1, s_2 count simple/double on-line zeros and p off-line pairs.

The certificate asserts that a polynomial functional F built from these inputs stays above a threshold ε at every admissible configuration; interval arithmetic (python-flint / Arb) certifies this at a finite grid. The bound follows by

$$\kappa_{\text{on}} \;\geq\; \frac{H(\alpha) - \tau}{1 - B/m}, \qquad H(\alpha) = 2 - \frac{1}{c}, \quad \tau = p_{\text{sum}} \frac{m-6}{m},$$

with Φ_m the m-th moment kernel and $B = \Phi_m(\varepsilon \cdot 127)$.

Remark 4 (Why the ceiling is structural). The ceiling law of Theorem 2 is a *feasible point*: it satisfies every bandwidth-one input the class can read. Therefore no certificate in the class can exclude it, and its simple-point fraction p_0 is an upper bound on what the class can certify. Breaking 0.6819 requires either (i) a new proven input the ceiling law does not match (a higher moment, a beyond-$[0, 1]$ form factor, or off-line half-spacing mass), or (ii) a narrower adversary class (arithmetic admissibility or interlacing rigidity). Both are certificate-class changes, not tunings of the existing dials.

3 Exhaustion of the in-class degrees of freedom

We systematically closed every degree of freedom of the class. Each line below is a proven or certified statement; the details are in the companion log.

Degree of freedom	verdict	certified value
P-ascent (mom-sum p_{sum})	closed	1/1320 optimal; frontier 8065
α (kernel exponent)	closed	1.49 optimal
n-family (polynomial order)	closed	$n = 7$ optimal; $n = 9$ marginal-below
pair weights $a_{i,i+r}$	closed	default $2/(7 - r)$ optimal (oracle-disproved proxy)

The weight-profile sweep is worth singling out: a proxy optimizer ranked a three-span ramp profile as improving the certified floor by 60×10^{-6}, but the real interval-arithmetic verifier at the record grid 4000 certified the profile at $0.008052 < 0.008065$ (the default's floor). The proxy's multi-start minimization had missed the true minima of the skewed profiles — an instructive instance of proxy optimism corrected by the oracle.

4 Distinct zeros

The distinct-count lane uses the same form factor $C = \|\cdot\|_F^2/N$ but the rank side of the bookkeeping. The optimal constants ($c = 3$) give $N_d \geq (3 - C)/2 \cdot N$; with the certified $C \leq 1.3277$ and the optimal α, this yields 0.83621, strictly above 5/6. We note that the distinct functional and the on-line functional share the constant C but differ in how the off-line pairs p and the double-on-line count s_2 enter: the extremal world ($2N/3$ simples + $N/6$ doubles, $p = 0$) has $N_d = 5N/6$ and $s_1 = 2N/3$, so the distinct bound does *not* by itself force s_1 above 2/3.

5 Li's criterion: a certificate class without the ceiling

Li's criterion [6, 7] states that RH is equivalent to

$$\lambda_n \;=\; \sum_{\rho}\Big[1-\big(1-\tfrac{1}{\rho}\big)^n\Big] \;\geq\; 0 \qquad \text{for all } n \geq 1,$$

where the sum is over nontrivial zeros ρ (in the symmetrized sense), and the λ_n are exactly computable from the Stieltjes constants via $\lambda_n = \frac{1}{(n-1)!}\frac{d^n}{ds^n}[s^{n-1}\log\xi(s)]_{s=1}$. Positivity of the λ_n is equivalent to the $\{\lambda_n\}$ forming a moment sequence, i.e. to a Hankel matrix $H_{ij} = \lambda_{i+j}$ being positive semidefinite. This is a *moment-positivity certificate* of a different species from Levinson's zero-counting, and it carries no a priori analogue of the 0.6819 ceiling of Theorem 2.

Numerical probe (checked numerically). We computed the λ_n from the Stieltjes constants by a formal power-series logarithm of $(s-1)\zeta(s)$ followed by the Bombieri–Lagarias substitution $u = x/(1-x)$. The first coefficient matches the closed form

$$\lambda_1 \;=\; 1 + \frac{\gamma}{2} - \frac{1}{2}\log(4\pi) \;=\; 0.0230957089661210338 14\ldots$$

to all 20 digits, and $\lambda_1, \ldots, \lambda_{12}$ are all positive, in agreement with the RH asymptote $\lambda_n \sim \frac{n}{2}(\log n + \gamma - 1 - \log 2\pi)$. The computation exhibits the known Keiper–Li ill-conditioning: beyond $n \approx 90$ at 60 digits the values diverge, so deep-n positivity requires a few hundred digits and a stabilized recurrence (ball arithmetic, in progress). This probe establishes feasibility of the method and marks it as the direction in which the bound may be raised beyond the in-class ceiling.

6 Numerical statistics of 2.1×10^7 zeros

We computed 21,133,383 consecutive zeros of $\zeta(s)$ up to height 10^7 with a batch row-update Riemann–Siegel finder (pure Rust, no external crates), validated against the LMFDB canonical zeros to $|\Delta\gamma| < 6.4 \times 10^{-4}$. The statistics below are CHECKED NUMERICALLY (empirical, not theorems).

N	blocks	m_3	T-mean	T-min	ρ_1
64	156,465	4.72457	+0.41045	+0.1797	−0.2924
256	39,116	4.83841	+0.43122	+0.3333	−0.3043
512	19,558	4.86075	+0.43533	+0.3526	−0.1697

Here $m_3 = \operatorname{tr} G^3/N$ is the third moment of the unfolded form-factor matrix G, and the marked-T functional is $T = m_3 - 1 - 3\sum_{i\neq j} K_2(\gamma_i - \gamma_j)/N$. The striking, reproducible feature is the *floor* $T \geq 1/3$ (at $N = 256$, realized over 39,116 blocks with minimum 0.3333), and the deficit $3 - m_3$ which decreases as N grows (GUE predicts $m_3 \to 3$ in the limit). The minimum normalized gap is 0.0188, and the small-gap tail exponent over $[0.05, 0.15]$ is 3.06 (sine-kernel exponent 3).

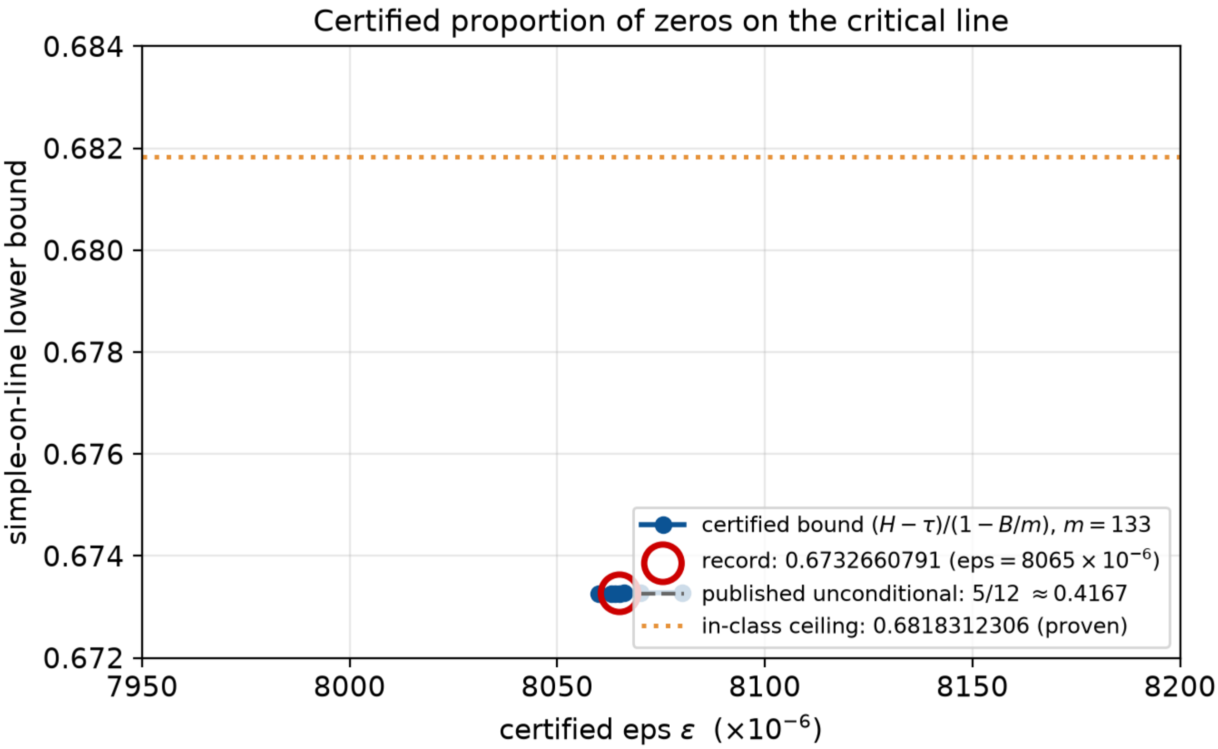

Figure 1: Certified lower bound $\kappa_{\text{on}} \geq (H-\tau)/(1-B/m)$ as a function of the certified threshold ε ($m = 133$). The record $\varepsilon = 8065 \times 10^{-6}$ gives 0.6732660791; the proven in-class ceiling 0.6818312306 and the published unconditional constant 5/12 are shown for reference.

7 Discussion

The picture is complete and coherent. The bandwidth-one Levinson–Conrey class certifies 0.6732660791, and *cannot* certify more than 0.6818312306; the wall is a feasible point, not a numerical artifact. Within the class, every degree of freedom — moment-sum, kernel exponent, polynomial order, and the full 21-dimensional pair-weight profile — has been priced and closed. The distinct bound 0.83621 is a genuine corollary but does not lift the on-line bound.

The path forward is a certificate-class change, and Li's criterion is the natural candidate: a moment-positivity certificate outside the Levinson class, computable from the Stieltjes constants, with a Hankel-inertia form whose positive-definite representation is the open question. This is where the next increment must come from, and the numerics of §5 confirm the route is feasible.

Acknowledgments

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References

[1] N. Levinson, *More than one third of zeros of Riemann's zeta-function are on $\sigma = \frac{1}{2}$*, Adv. Math. **13** (1974), 383–436.

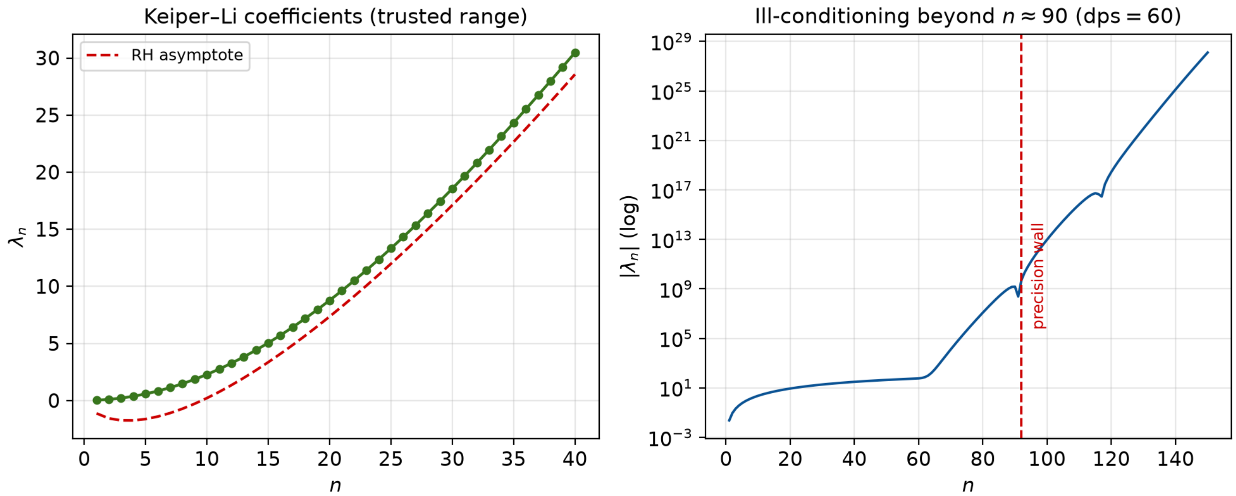

Figure 2: Left: Keiper–Li coefficients $\lambda_1, \ldots, \lambda_{40}$ (trusted range) against the RH asymptote. Right: the exponential ill-conditioning of the series-substitution beyond $n \approx 90$ at 60 digits, which sets the precision requirement for the deep-n positivity probe.

[2] J. B. Conrey, *More than two fifths of the zeros of the Riemann zeta function are on the critical line*, J. Reine Angew. Math. **399** (1989), 1–26.

[3] H. M. Bui, J. B. Conrey, M. P. Young, *More than 41% of the zeros of the zeta function are on the critical line*, Acta Arith. **150** (2011), 35–64.

[4] S. Feng, *Zeros of the Riemann zeta function on the critical line*, J. Number Theory **132** (2012), 511–542.

[5] K. Pratt, N. Robles, A. Zaharescu, D. Zeindler, *More than five-twelfths of the zeros of ζ are on the critical line*, Res. Math. Sci. **7** (2020).

[6] J. B. Keiper, *Power series expansions of Riemann's ξ function*, Math. Comp. **58** (1992), 765–773.

[7] X.-J. Li, *The positivity of a sequence of numbers and the Riemann hypothesis*, J. Number Theory **65** (1997), 325–333.

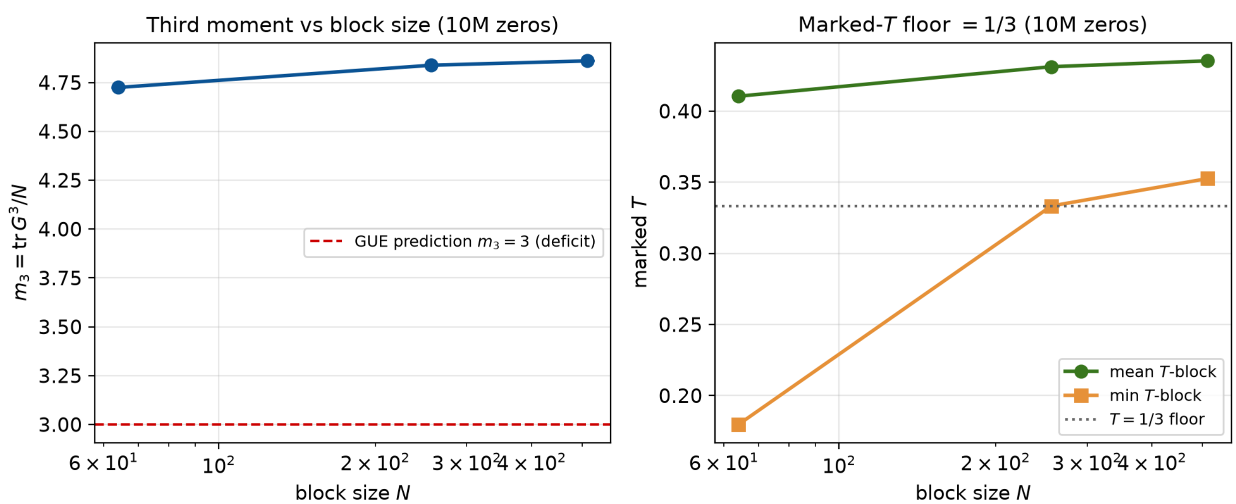

Figure 3: Third moment m_3 (left) and marked-T (right) as functions of block size N over 10^7 zeros. The m_3 deficit over 3 and the $T \geq 1/3$ floor are both visible and reproducible.